

SHENGYANG WU

syseanwu@umich.edu

1037 Beal Ave ◇ Ann Arbor, MI 48105

EDUCATION

University of Michigan

Ph.D. in Electrical Engineering

Supervisor: Prof. Vladimir Dvorkin

Ann Arbor, MI, US

September 2024 - May 2028

GPA:3.7

Huazhong University of Science and Technology

M.Sc. in Electrical Engineering

Supervisor: Prof. Dongyuan Shi

Wuhan, China

September 2021 - July 2024

GPA:4.0

North China Electric Power University

B.Sc. in Electrical Engineering and its Automation

Supervisor: Prof. Zhaohao Ding

Beijing, China

September 2017 - July 2021

GPA:3.96

RESEARCH AREA

Power system optimization, differential privacy, electric vehicles, power markets

WORK EXPERIENCE

Grid Intelligence Intern - Eaton Research Lab

May 2026 - August 2026

- Designed a Graph Neural Network (GNN)-based module in Eaton's Feeder Automation Manager (FAM) for Volt-VAR control under limited measurement and inaccurate parameters in distribution system control.
- Implemented the Volt-VAR control strategies in a real distribution feeder. Reduced the occurrence of voltage violation by 50% compared to existing controllers.

RESEARCH PROJECT

Monetizing Electrical Vehicle for Electricity Market

September 2025 - Present

- Developed online EV charging coordination strategies under market signals.
- Modeled the privacy guarantee for individual EV users in online feedback algorithms.

Differential Privacy for Power System Dynamics

April 2025 - Present

- Developed an algorithm for the private release of dynamic system models.
- Implemented the algorithm for power system dynamics and achieved high model fidelity while guaranteeing the privacy of power system parameters.

Cyber-Resilient Synthetic Data Generation

September 2024 - March 2025

- Developed a differentially private algorithm for power system operators to release sensitive data.
- Establish guarantees that the released data cannot be abused for launching cyber attacks.

PUBLICATIONS

Journal Publications

- J1 **S. Wu.** and V. Dvorkin. Synthesizing Grid Data with Cyber Resilience and Privacy Guarantees. *IEEE Control System Letters*, vol.9, pp.438-443, 2025. doi:10.1109/LCSYS.2025.3574146
- J2 C. Yang, **S. Wu.**, T. Liu, et al. Efficient Generation of Power System Topology Diagrams Based on Graph Neural Network. *Engineering Applications of Artificial Intelligence*, 2025, 149: 110462. doi:10.1016/j.engappai.2025.110462

- J3 **S. Wu**, C. Yang, D. Shi and J.Wang. A Lightweight Framework for Measurement Causality Extraction and FDIA Localization. *IEEE Transactions on Smart Grid*, vol.16, no.3, pp.2587-2598, May 2025, doi: 10.1109/TSG.2025.3548097.
- J4 D. Hu, **S. Wu**, J. Wang, et al. Training a Dynamic Neural Network to Detect False Data Injection Attacks Under Multiple Unforeseen Operating Conditions. *IEEE Transactions on Smart Grid*, vol. 15, no. 3, pp. 3248-3261, May 2024, doi: 10.1109/TSG.2023.3339975.
- J5 **S. Wu**, Z. Ding , J. Wang , et al. Unveiling bidding uncertainties in electricity markets: A Bayesian deep learning framework based on accurate variational inference. *Energy*, 2023, 276: 127286. doi: 10.1016/j.energy.2023.127286.

Conference Publications

- C1 Z. Ding, X. Zheng, C. Liao, Y. Lyu, J. Chen, **S. Wu**, L. Zhang, and X. Hu. "Omnisam: Omnidirectional segment anything model for uda in panoramic semantic segmentation." In Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 23892-23901. 2025.
- C2 **S. Wu**, D. Hu, Y. Gao, J. Wang and D. Shi, "Interpretable Detection and Localization of False Data Injection Attacks Based on Causal Learning", *2023 IEEE Power & Energy Society General Meeting (PESGM)*, Orlando, FL, USA, 2023, pp. 1-5, doi: 10.1109/PESGM52003.2023.10253166.
- C3 Z. Zhao, Y. Zou, **S. Wu**, et al, "VSC-HVDC Traveling Wave Protection Based on the First Current Extremum of the Non-Fault Pole", *2023 IEEE Power & Energy Society General Meeting (PESGM)*, Orlando, FL, USA, 2023, pp. 1-5, doi: 10.1109/PESGM52003.2023.10252595.

REVIEWER SERVICES

IEEE Transactions on Power Systems	<i>Since 2025</i>
IEEE Transactions on Smart Grids	<i>Since 2024</i>
IEEE Transactions on Energy Markets, Policy, and Regulation	<i>Since 2024</i>
Journal of Modern Power Systems and Clean Energy	<i>Since 2024</i>
IEEE Power and Energy Society General Meeting	<i>Since 2023</i>

AWARDS

National Scholarships of China	<i>2018 & 2019</i>
Gerstacker Fellowship	<i>2024-2025</i>
Gold Award of 2026 EVC Symposium	<i>2026</i>